

Taiyu Zhang

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SUMMARY

PhD candidate at KU Leuven Computer Science. Hands-on across the language-model lifecycle: from-scratch pretraining (40 GB cleaned Chinese corpus, GPT-2 scale, shipped in the iFlyTek input method), large-model distillation (LingBot video diffusion, 50 min → < 4 min), multimodal data pipelines (diffusion generation + VLM annotation), and LLM-agent systems (IVA 2024 first author; 1.3k+ GitHub stars).

EXPERIENCE

- Research Intern, Multimodal Interaction — ByteDance PICOJul 2026 – Present
 - Research on the synthetic-data pipeline for hand-gesture tracking on XR devices.
- Research Intern — Moxin TechnologyMar 2026 – May 2026
 - Autoregressive training and acceleration of long-video generation models.
- NLP Researcher — iFlyTek2021 – 2022
 - Multiple national-level R&D projects; main owner of the multilingual semantic-understanding component of the iFlyTek Digital Human — data curation, model training, and evaluation.
 - Deployed in the 2022 Beijing Winter Olympics and other state-level events.
 - Long-sentence-completion GPT pretraining for the iFlyTek input method.

SELECTED PROJECTS

- Hand-Gesture Data Pipeline — Diffusion data generation + multimodal VLM annotation2026.07 – Present
 - Diffusion-based generation of hand-gesture video data, widening training coverage past what captured sessions reach: unusual hand shapes, lighting, skin tones, viewing angles, and occlusion.
 - Annotation of the gesture video data with the QwenVL multimodal understanding model, together with review of annotation quality.
- LingBot Distillation — 14B video diffusion → 4-step student on Ascend2026.03 – 2026.05
 - Reproduced LingBot’s distillation on Huawei Ascend (CANN); teacher: camera-trajectory control + 1000-frame long-video coverage. 40–50-step teacher → 4-step student; > 50-min generation → < 4 min.
 - Resolved self-forcing memory bottleneck with two-pass forward + activation recompute (gradient-accumulation style); proposed a new ODE-initialization that uses training data to init student directly from teacher, removing explicit ODE-pair generation and stabilising distillation.
- Claw-AI-Lab — Co-developer2025 – Present
 - Multi-agent autonomous research platform: one prompt → paper, code, figures, and experiment logs. 1.3k+ GitHub stars ([Claw-AI-Lab/Claw-AI-Lab](#)).
 - Contributed across the multi-agent scheduler, the local-execution harness, and the real-time monitoring dashboard.
- iFlyTek IME Long-Sentence Completion — GPT pretraining from scratch2021 – 2022
 - Built a 400 GB → 40 GB pretraining-data pipeline: deduplication, language and quality filtering, topic balancing, formatting normalization.
 - Pretrained a GPT-style language model from scratch on the cleaned 40 GB Chinese corpus.
 - Shipped in the iFlyTek input method, powering long-text completion and poetry-creation features in production.
- PhD Research: LLM Agents in Immersive Environments — First author on all three2023 – Present
 - Focus Agent (IVA 2024) — LLM-driven virtual focus group: each participant is an agent with persona, reasoning style, and memory; a moderator steers the session toward an interview goal.
 - AI Game NPC (SUI 2026) — LLM companion in a VR puzzle game: Whisper perception, GPT-4 cognition querying game-engine state directly, TTS-plus-action executor; 24-participant study across four assistance levels.
 - Coembodiment AI — human and AI sharing control of one virtual body; a non-linear adaptive control function outperforms gate / linear baselines.

EDUCATION

- Ph.D., Computer Science — KU Leuven2022 – Present

Intelligent virtual agents in 3D environments. Advisor: Prof. Adalberto L. Simeone.
- M.S., Electronic Engineering — Tsinghua University2018 – 2021

Advisor: Prof. Xiaohong Ma.
- B.S., Electronic Engineering & Information Science — USTC2014 – 2018

Advisor: Prof. Jian Tang.

SELECTED PUBLICATIONS

- MoWorld: A Flash World Model.**
Team Moxin, incl. Zhang, T. Technical report, [arXiv:2607.06216](#), 2026.
- “What Can I Do for You”: How Should AI Companions Provide Assistance to Players in Virtual Reality Games.**
Zhang, T., et al. *ACM Symposium on Spatial User Interaction (SUI)*, 2026.
- Focus Agent: LLM-Powered Virtual Focus Group.**
Zhang, T., et al. *Proceedings of the 24th ACM International Conference on Intelligent Virtual Agents (IVA)*, 2024. [arXiv:2409.01907](#).

SKILLS

- LanguagesPython · C++ · Swift · TypeScript · C#
- ML / SystemsPyTorch · Pretraining · Distillation · Tokenization (VQ-VAE / VQGAN) · Distributed training · Ascend / CANN
- ModelingTransformer / GPT · Diffusion · Autoregressive sequence modeling · Multi-agent orchestration
- SpokenEnglish (fluent) · Chinese (native)